

Aeroservoelastic Modeling and Analysis of a Missile Control Surface with a Nonlinear Electromechanical Actuator

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In this study, aeroservoelastic modeling and analysis of a missile control surface which is operated and controlled by a power limited, nonlinear electromechanical actuator is performed. Linear models of the control fin structure and aerodynamics together with the nonlinear servo-actuation system are built and integrated. The resulting aeroservoelastic system is analyzed both in time and frequency domain. Structural model of the control fin is based on the finite element model of the fin. Aerodynamic model of the control fin is based on the Generalized Aerodynamic Force (GAF) Matrix model which is generated in accordance with the finite element model of the fin. In order to be able to represent the aerodynamic forces in the continuous frequency domain, and the time domain, a rational function approximation formulation of the GAF matrices is utilized. Both the elastic and the rigid body motion of the missile control fin are modeled through modal discretization, so that the interaction between the aeroelastic system and the control system dynamics can be treated appropriately. A PD controller synthesis is carried out using the Root Locus Method, by neglecting the flexibility of the fin and unsteady aerodynamic effects on the fin, as if the fin is rigid and the aerodynamics is steady. The shortcomings of these assumptions are discussed under the effect of aeroelastic loading. A third order servoactuation system together with the PD controller is integrated to the aeroelastic system, so that the limit cycle oscillations of the aeroservoelastic system due to power limit saturation could be studied. Two alternative, common position feedback paths for the PD controller are used and the resulting aeroservoelastic systems are simulated in response to various control commands. It is shown that the stability and dynamic performance of the aeroservoelastic system also depends on the selection of the feedback path, when aeroelastic dynamics is present.

I. Introduction

The vast majority of problems regarding the flexibility of missiles are focused on control surfaces such as fins or canards, since the slenderness of these control surfaces are more pronounced than that of the missile body. Moreover, the interaction of structural dynamics with control system dynamics renders the control surface design and analysis more intriguing due to the frequency separation between structural and control system dynamics being relatively narrow and the motion feedback sensors having high enough bandwidth to measure flexible motion. On the contrary, the motion due to flexible modes of the missile body are usually naturally filtered by the bandwidth of onboard inertial measurement units and the control command generating autopilots during flight, resulting in less interaction with the overall missile control system.

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The traditional way of modeling the flight mechanics of an aircraft is followed by making the assumption that the structure is rigid throughout the motion. As a result, dynamics of the aircraft is modeled only by accounting for the aerodynamic and the inertial forces. This type of modeling is most frequently used for the aircraft control system design and flight performance evaluation purposes. The validity of a rigid body model is questionable when the structure undergoes significant elastic motion, which in turn, alters the aerodynamic forcing on the aircraft. A dynamic aeroelastic formulation involving the control system dynamics is necessary for modeling, analyzing and performing control design when the plant in consideration is a missile control fin with significant elastodynamics.

Control actuation systems of missiles are of various types. Electro-hydraulic, pneumatic and electro-mechanical control actuation systems exist, and each type has a distinctive major impact on the dynamic behavior of the aeroservoelastic system. Electromechanical actuation systems are widely preferred because of their conservative power consumption, linearity, denser packaging, and good performance at disturbance load rejection.

The structural dynamics, aerodynamics and the actuator dynamics have long been represented by linear models, rendering possible the analysis of the aeroservoelastic system in the frequency domain¹⁻³. The structural dynamics of all movable control surfaces with linear structural properties is commonly modeled utilizing the finite element method, putting forward modal discretization to account for both the rigid body and the flexible motion⁴⁻⁶. Together with the finite element formulation of the structural dynamics, the Doublet Lattice Method⁷ of aerodynamics has been used for carrying out flutter analysis and obtaining the Generalized Aerodynamic Force (GAF) matrices at discrete reduced frequencies. These GAF matrices are then input to a Rational Function Approximation (RFA) method such as Roger's RFA⁸ to obtain continuous frequency domain representation of aerodynamics over a specified frequency range⁹⁻¹⁰ as a function of the Laplace variable 's'. The inverse Laplace transform of the continuous frequency domain representation renders possible the time domain analysis of the aeroservoelastic system¹¹.

The present study presents a generic aeroservoelastic model, that renders possible the important physical phenomena involved in the aeroservoelastic motion to be exploited. To provide genericity in structural modeling, the finite element formulation of structural dynamics is utilized. Through the use of the finite element formulation, implementation of various boundary conditions and various geometries are made possible. One way of using the practical fictitious mass concept⁶ is also presented. Aerodynamic modeling is handled with intermediate complexity, by utilizing panel discretization formulation of the Doublet Lattice Method. The aerodynamic forces are modeled with Generalized Aerodynamic Force matrices, to introduce additional modularity, since the extraction of these matrices is also possible with higher fidelity CFD solvers. The Generalized Aerodynamic Force matrices are fitted for a range of reduced frequencies with Roger's RFA⁸ method, so that the oscillatory aerodynamic forces can be represented in the time domain. The aerodynamic and structural matrices are integrated, also accounting for rigid body motion, through a procedure which is well documented by Karpel². The formulation allowed the state space representation of the dynamics of aeroservoelasticity. Flutter analysis and normal modes analysis of the models built are carried out both in MSC Nastran and MATLAB for verification. It is observed that the results of analysis carried out in MSC Nastran and MATLAB are very close, even if different methods such as the p-k method, root locus method and time domain simulation is utilized for flutter analysis. The scope of the analysis of the aeroservoelastic model in frequency and time domain are expanded to linear analysis, power limited nonlinear analysis, stability analysis with different feedback paths and performance analysis with regard to performance requirements.

II. Aeroelastic Modeling

A. Aeroelastic Mathematical Model

In aircraft structural dynamics analysis, modal reduction is usually employed to reduce the problem size. The physical motivation behind the reduction process is that, in general, a relatively narrow frequency range of interest dominates the dynamics of the structure. Through a proper selection of a number of low frequency structural modes, it is possible to capture the dominant dynamics of a structure and reduce the problem size. For the cases where an independent external forcing is also involved, the frequency content of the external forcing agents should be taken into account. Equations (1) and (2) represent the equation of motion of the missile control fin with respect to modal coordinates ζ_m , in time domain and Laplace domain respectively.

$$[\mathbf{M}_{mm}]\{\ddot{\zeta}_m(t)\} + [\mathbf{C}_{mm}]\{\dot{\zeta}_m(t)\} + [\mathbf{K}_{mm}]\{\zeta_m(t)\} = \{\mathbf{F}_m(\zeta_m, \dot{\zeta}_m, \ddot{\zeta}_m, t)\} \quad (1)$$

$$(s^2 [\mathbf{M}_{mm}] + s [\mathbf{C}_{mm}] + [\mathbf{K}_{mm}]) \{\zeta_m(s)\} = \{F_m(s)\} \quad (2)$$

In Eq. (2), $\mathbf{M}_{mm}$, $\mathbf{C}_{mm}$ and $\mathbf{K}_{mm}$ are the reduced modal mass, damping and stiffness matrices respectively, and F_m is the vector of external forces at modal coordinates. Equation (2) can be rewritten, by expressing the aeroelastic system at the left hand side, and the external control force applied by the actuator at the right hand side of Eq. (3).

$$(s^2 [\mathbf{M}_{mm}] + s [\mathbf{C}_{mm}] + [\mathbf{K}_{mm}] - q[\bar{\mathbf{Q}}_{mm}(M, s)]) \{\zeta_m(s)\} = \{T_{act}(s)\} \quad (3)$$

In Eq. (3) q is the dynamic pressure, $T_{act}(s)$ is the moment applied by the actuator to the fin and $\bar{\mathbf{Q}}_{mm}$ is the reduced modal aerodynamic forcing matrix. Equation (3) is partitioned with respect to rigid body and flexible modal coordinates to obtain Eq. (4).

$$\left(s^2 \begin{bmatrix} \mathbf{M}_{cc} & \mathbf{M}_{cs} \\ \mathbf{M}_{sc} & \mathbf{M}_{ss} \end{bmatrix} + s \begin{bmatrix} 0 & 0 \\ 0 & \mathbf{C}_{ss} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \mathbf{K}_{ss} \end{bmatrix} - q \begin{bmatrix} \bar{\mathbf{Q}}_{cc} & \bar{\mathbf{Q}}_{cs} \\ \bar{\mathbf{Q}}_{sc} & \bar{\mathbf{Q}}_{ss} \end{bmatrix} \right) \begin{Bmatrix} \zeta_c(s) \\ \zeta_s(s) \end{Bmatrix} = \begin{Bmatrix} T_{fin}(s) \\ 0 \end{Bmatrix} \quad (4)$$

Equation (4) can be rewritten in two equation sets as follows :

$$(s^2 \mathbf{M}_{cs} - q \bar{\mathbf{Q}}_{cs}) \zeta_s(s) + (s^2 \mathbf{M}_{cc} - q \bar{\mathbf{Q}}_{cc}) \zeta_c(s) = T_{fin}(s) \quad (5)$$

$$(s^2 \mathbf{M}_{ss} + s \mathbf{C}_{ss} + \mathbf{K}_{ss} - q \bar{\mathbf{Q}}_{ss}) \zeta_s(s) + (s^2 \mathbf{M}_{sc} - q \bar{\mathbf{Q}}_{sc}) \zeta_c(s) = 0 \quad (6)$$

In Eq.(5), the coupling between rigid body dynamics with aeroelastic dynamics is introduced by the inertial and the aerodynamic coupling on $\zeta_s(s)$, which is the elastic modal coordinate vector. In Eq.(6), the coupling between rigid body dynamics with aeroelastic dynamics is introduced by inertial and aerodynamic coupling on $\zeta_c(s)$, the rigid body modal coordinate, which is the actuator output $\theta_{fin}(s)$. The total hinge moment acting on the transmission unit output shaft, including the aerodynamic and the inertial components is given by Eq. (7).

$$T_{hinge@fin}(s) = -T_{fin}(s) \quad (7)$$

The aerodynamic force matrices in Eq. (5) and Eq. (6) are obtained from Roger's RFA utilizing Eq. (8), where the relation between the reduced frequency and the Laplace variable is taken as $ik = sb/U$. In Eq. (8), $\mathbf{A}_{0...N+2}$ are the constant matrices to be determined from least square approximation for the best fit of $\mathbf{Q}(M, s)$, which is obtained using the DLM. β_n are the aerodynamic lag roots which are preset to specific values in the frequency range of interest, so that the aerodynamic lag functions are stable, the least square problem becomes linear, and the fit is satisfactory. A detailed formulation of Roger's RFA is given in the next sub-subsection.

$$[\bar{\mathbf{Q}}(M, s)] = [\mathbf{A}_0] + \frac{sb}{U} [\mathbf{A}_1] + \left(\frac{sb}{U} \right)^2 [\mathbf{A}_2] + \sum_{n=1}^N \frac{s[\mathbf{A}_{n+2}]}{s + \frac{b}{\beta_n}} \quad (8)$$

Aeroelastic matrices given below are defined for subsequent use. In the definition of the aeroelastic matrices, all coefficients $\mathbf{A}_{0...N+2}$ are obtained from Roger's RFA.

$$\begin{aligned} \bar{\mathbf{M}}_{cs} &= \mathbf{M}_{cs} - \frac{\rho b^2}{2} \mathbf{A}_{cs2} & \bar{\mathbf{M}}_{cc} &= \mathbf{M}_{cc} - \frac{\rho b^2}{2} \mathbf{A}_{cc2} & \bar{\mathbf{K}}_{cs} &= -\frac{\rho U^2}{2} \mathbf{A}_{cs0} & \bar{\mathbf{K}}_{cc} &= -\frac{\rho U^2}{2} \mathbf{A}_{cc0} \\ \bar{\mathbf{C}}_{cs} &= -\frac{\rho b U}{2} \mathbf{A}_{cs1} & \bar{\mathbf{C}}_{cc} &= -\frac{\rho b U}{2} \mathbf{A}_{cc1} & \bar{\mathbf{M}}_{ss} &= \mathbf{M}_{ss} - \frac{\rho b^2}{2} \mathbf{A}_{ss2} & \bar{\mathbf{M}}_{sc} &= \mathbf{M}_{sc} - \frac{\rho b^2}{2} \mathbf{A}_{sc2} \\ \bar{\mathbf{C}}_{ss} &= \mathbf{C}_{ss} - \frac{\rho b U}{2} \mathbf{A}_{ss1} & \bar{\mathbf{C}}_{sc} &= -\frac{\rho b U}{2} \mathbf{A}_{sc1} & \bar{\mathbf{K}}_{ss} &= \mathbf{K}_{ss} - \frac{\rho U^2}{2} \mathbf{A}_{ss0} & \bar{\mathbf{K}}_{sc} &= -\frac{\rho U^2}{2} \mathbf{A}_{sc0} \end{aligned}$$

Two types of aerodynamic lag states are defined so that a state space formulation for the solution of $\zeta_s(s)$ could be constructed. The aeroelastic hinge moment lag state is given by Eq. (9), which is used for constructing the hinge moment as an output of the state space aeroelastic system model. The lag state for the aeroelastic equation of motion is given in Eq. (10).

$$\zeta_{can}(s) = \frac{s\mathbf{A}_{csn+2}}{s + \frac{U}{b}\beta_n} \zeta_s(s) + \frac{s\mathbf{A}_{ccn+2}}{s + \frac{U}{b}\beta_n} \zeta_c(s) \quad (9)$$

$$\zeta_{san}(s) = \frac{s\mathbf{A}_{ssn+2}}{s + \frac{U}{b}\beta_n} \zeta_s(s) + \frac{s\mathbf{A}_{scn+2}}{s + \frac{U}{b}\beta_n} \zeta_c(s) \quad (10)$$

Then, the state space formulation of the aeroelastic system augmented with hinge moment lag states is obtained as in Eq. (11).

$$\{\dot{z}(t)\} = [\mathbf{A}_{sys}] \{z(t)\} + [\mathbf{B}_{sys}] \{u(t)\} \quad (11)$$

The system and input matrices are expressed as:

$$[\mathbf{A}_{sys}] = \begin{bmatrix} 0 & \mathbf{I} & 0 & 0 \\ -\bar{\mathbf{M}}_{ss}^{-1}\bar{\mathbf{K}}_{ss} & -\bar{\mathbf{M}}_{ss}^{-1}\bar{\mathbf{C}}_{ss} & q\bar{\mathbf{M}}_{ss}^{-1}\mathbf{D}_s & 0 \\ 0 & \mathbf{E}_{ss} & \frac{U}{b}\mathbf{R}_{ss} & 0 \\ 0 & \mathbf{E}_{cs} & 0 & \frac{U}{b}\mathbf{R}_{cs} \end{bmatrix} \quad \{z(t)\} = \begin{Bmatrix} \zeta_s(t) \\ \dot{\zeta}_s(t) \\ \zeta_{san}(t) \\ \dots \\ \zeta_{san}(t) \\ \zeta_{can}(t) \\ \dots \\ \zeta_{can}(t) \end{Bmatrix}$$

$$[\mathbf{B}_{sys}] = \begin{bmatrix} 0 & \mathbf{I} & 0 \\ \bar{\mathbf{M}}_{ss}^{-1}\bar{\mathbf{K}}_{sc} & \bar{\mathbf{M}}_{ss}^{-1}\bar{\mathbf{C}}_{sc} & -\bar{\mathbf{M}}_{ss}^{-1}\bar{\mathbf{M}}_{sc} \\ 0 & \mathbf{E}_{sc} & 0 \\ 0 & \mathbf{E}_{cc} & 0 \end{bmatrix} \quad \{u(t)\} = \begin{Bmatrix} \zeta_c(t) \\ \dot{\zeta}_c(t) \\ \ddot{\zeta}_c(t) \end{Bmatrix}$$

where zeros and $\mathbf{I}$'s represent zero and identity matrices, z is the vector representing the states including the aerodynamic lag states, and u is the vector representing the control input.

$$\mathbf{E}_{ss} = \begin{Bmatrix} \mathbf{A}_{ss3} \\ \mathbf{A}_{ss4} \\ \dots \\ \mathbf{A}_{ssN+2} \end{Bmatrix} \quad \mathbf{E}_{sc} = \begin{Bmatrix} \mathbf{A}_{sc3} \\ \mathbf{A}_{sc4} \\ \dots \\ \mathbf{A}_{scN+2} \end{Bmatrix} \quad \mathbf{R}_{ss} = \begin{bmatrix} -\beta_1 & 0 & 0 & 0 \\ 0 & -\beta_2 & 0 & 0 \\ 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & -\beta_N \end{bmatrix}$$

$$\mathbf{E}_{cs} = \begin{Bmatrix} \mathbf{A}_{cs3} \\ \mathbf{A}_{cs4} \\ \dots \\ \mathbf{A}_{csN+2} \end{Bmatrix} \quad \mathbf{E}_{cc} = \begin{Bmatrix} \mathbf{A}_{cc3} \\ \mathbf{A}_{cc4} \\ \dots \\ \mathbf{A}_{ccN+2} \end{Bmatrix} \quad \mathbf{R}_{cs} = \mathbf{R}_{ss} = \begin{bmatrix} -\beta_1 & 0 & 0 & 0 \\ 0 & -\beta_2 & 0 & 0 \\ 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & -\beta_N \end{bmatrix}$$

$$\mathbf{D}_s = [\mathbf{I}_1 \quad \mathbf{I}_2 \quad \dots \quad \mathbf{I}_N]$$

In the relations given above, $\mathbf{D}_s$ is composed of N identity matrices of size equal to the number of elastic modal coordinates. The output of the aeroelastic state space system model is formulated, so that the actual displacements at global coordinates, and the hinge moment acting on the fin is calculated. In order to obtain global displacements, modal superposition is used as in Eq. (12).

$$x(t) = [\phi_{as}] \{\zeta_s(t)\} + [\phi_{ac}] \{\zeta_c(t)\} \quad (12)$$

Hinge moment is expressed in terms of states z and inputs u by Eq.(13)

$$T_{hinge @ fin}(s) = T_{hinge @ fin}^{state}(s) + T_{hinge @ fin}^{input}(s) \quad (13)$$

where,

$$T_{hinge @ fin}^{state}(s) = (-s^2 \bar{\mathbf{M}}_{cs} - s \bar{\mathbf{C}}_{cs} - \bar{\mathbf{K}}_{cs} + q x_{cs}^l(s)) \zeta_s(s) \quad (14)$$

$$T_{hinge @ fin}^{input}(s) = (-s^2 \bar{\mathbf{M}}_{cc} - s \bar{\mathbf{C}}_{cc} - \bar{\mathbf{K}}_{cc} + q x_{cc}^l(s)) \zeta_c(s) \quad (15)$$

where cc , cs , sc and ss sets of $x^l(s)$ is given by $x^l(s) = \sum_{n=1}^N \frac{s \mathbf{A}_{n+2}}{s + \frac{U}{b} \beta_n}$.

Formulating Eq. (14) and Eq. (15) in state space form, the hinge moment is formulated as an output in Eq. (16).

$$T_{hinge @ fin}(t) = [\mathbf{C}_{hinge @ fin}] \{z(t)\} + [\mathbf{D}_{hinge @ fin}] \{u(t)\} \quad (16)$$

The output matrix and the direct transition matrix are defined in Eq. (17) and Eq. (18),

$$\mathbf{C}_{hinge @ fin} = [-\bar{\mathbf{K}}_{cs} \quad -\bar{\mathbf{C}}_{cs} \quad 0 \quad q \mathbf{D}_c] - \bar{\mathbf{M}}_{cs} \bar{\mathbf{M}}_{ss}^{-1} [-\bar{\mathbf{K}}_{ss} \quad -\bar{\mathbf{C}}_{ss} \quad q \mathbf{D}_s \quad 0] \quad (17)$$

$$\mathbf{D}_{hinge @ fin} = [-\bar{\mathbf{K}}_{cc} \quad -\bar{\mathbf{C}}_{cc} \quad -\bar{\mathbf{M}}_{cc}] - \bar{\mathbf{M}}_{cs} \bar{\mathbf{M}}_{ss}^{-1} [\bar{\mathbf{K}}_{sc} \quad \bar{\mathbf{C}}_{sc} \quad -\bar{\mathbf{M}}_{sc}] \quad (18)$$

where $\mathbf{D}_c$ is an N size row vector composed of 1's. Eq. (11), Eq. (12) and Eq. (16) form the state space aeroelastic system model together with its outputs as displacements in global coordinates and total hinge moment around the fin. All the aeroelastic matrices presented in the formulation are obtained by normal modes and flutter analyses carried out in MSC Nastran.

1. Rational Function Approximation to Unsteady Aerodynamics

Aeroservoelastic formulation of the missile control fin requires transient aerodynamic effects to be modeled. The GAF matrices obtained from DLM cannot be directly used to model transient aerodynamic phenomena, since DLM is only directly applicable to steady state sinusoidal motion. All RFA methods depend on least square techniques, which fit a frequency response function to GAF matrices at multiple reduced frequencies. Comparisons of accuracy and computational efficiency between these methods are available in Ref. 1. Roger's Method approximates the unsteady aerodynamics with following frequency response function that depends on reduced frequency and Mach number:

$$[\bar{\mathbf{Q}}(M, k)] = [\mathbf{A}_0] + (ik)[\mathbf{A}_1] + (ik)^2[\mathbf{A}_2] + \sum_{n=1}^N \frac{(ik)[\mathbf{A}_{n+2}]}{ik + \beta_n} \quad (19)$$

In Eq. (19), $\mathbf{A}_{0..N+2}$ are the constant matrices to be determined from least square approximation for the best fit of $\mathbf{Q}(M, k)$. β_n are the aerodynamic lag roots. β_n are preset to specific values in the frequency range of interest, so that the aerodynamic lag functions are stable, the least square problem becomes linear, and the fit is satisfactory. Note that for each element of $\mathbf{Q}(M, k)$ at different frequencies $\mathbf{Q}^{ij}(M, k_1), \mathbf{Q}^{ij}(M, k_2) \dots \mathbf{Q}^{ij}(M, k_{n_k})$, a least square fit is generated. Given a $\mathbf{Q}(M, k)$ matrix of size n_m at a Mach number, the number of linear least square fits required is $n_m^2 \times n_k$, where n_k is the number of GAF matrices at n_k different reduced frequencies. β_n are the same for each

fit, so that the size of the resulting aeroelastic equation is kept small. By modeling the unsteady aerodynamics with Roger's Method, number of states of the resulting aeroelastic equation will be $n_s(N+2)$, where n_s is the number of elastic modal degrees of freedom. To calculate $\mathbf{A}_{0...N+2}^{ij}$ for a Mach number, $\bar{\mathbf{Q}}^{ij}(M, k)$ is divided into real and imaginary parts such that:

$$[\bar{\mathbf{Q}}^{ij}(M, k)] = [\bar{\mathbf{Q}}_R^{ij}(M, k)] + i[\bar{\mathbf{Q}}_I^{ij}(M, k)] \quad (20)$$

$$[\bar{\mathbf{Q}}_R^{ij}(M, k)] = [\mathbf{A}_{0}^{ij}] - k^2[\mathbf{A}_{2}^{ij}] + \sum_{n=1}^N \frac{k^2[\mathbf{A}_{n+2}^{ij}]}{k^2 + \beta_n^2} \quad (21)$$

$$[\bar{\mathbf{Q}}_I^{ij}(M, k)] = k[\mathbf{A}_{1}^{ij}] + \sum_{n=1}^N \frac{\beta_n k[\mathbf{A}_{n+2}^{ij}]}{k^2 + \beta_n^2} \quad (22)$$

Then these parts are written in matrix form, for each Mach - reduced frequency pair:

$$[\bar{\mathbf{Q}}_R^{ij}(M, k_f)] = \{K_{Rf}\}[\mathbf{A}^{ij}] \quad (23)$$

$$[\bar{\mathbf{Q}}_I^{ij}(M, k_f)] = \{K_{If}\}[\mathbf{A}^{ij}] \quad (24)$$

where K_{Rf} , K_{If} and $\mathbf{A}^{ij}$ are defined as:

$$K_{Rf} = \begin{bmatrix} 1 & 0 & -k_f^2 & \frac{k_f^2}{k_f^2 + \beta_1^2} & \frac{k_f^2}{k_f^2 + \beta_2^2} & \cdots & \frac{k_f^2}{k_f^2 + \beta_N^2} \end{bmatrix} \quad (25)$$

$$K_{If} = \begin{bmatrix} 0 & k_f & 0 & \frac{\beta_1 k_f}{k_f^2 + \beta_1^2} & \frac{\beta_2 k_f}{k_f^2 + \beta_2^2} & \cdots & \frac{\beta_N k_f}{k_f^2 + \beta_N^2} \end{bmatrix} \quad (26)$$

$$\mathbf{A}^{ij} = [\mathbf{A}_{0}^{ij} \quad \mathbf{A}_{1}^{ij} \quad \mathbf{A}_{2}^{ij} \quad \mathbf{A}_{3}^{ij} \cdots \mathbf{A}_{(N+2)}^{ij}]^T \quad (27)$$

Then the following complex error function is defined for linear least square minimization problem:

$$[\mathbf{E}^{ij}(M, k_f)] = [\mathbf{Q}^{ij}(M, k_f)] - [\bar{\mathbf{Q}}^{ij}(M, k_f)] \quad (28)$$

The aim is to make a linear least square fit, so that the left hand side of Eq. (28) is minimized. Then,

$$\frac{\partial}{\partial \mathbf{A}_{ij}^f} \sum_{n=1}^{n_k} (\mathbf{E}^{ij}(M, k_f) \mathbf{E}^{ij}(M, k_f)') = 0 \quad (29)$$

Eq. (29) is solved to obtain Eq. (30).

$$\mathbf{A}^{ij} = \left[\sum_{f=1}^{n_k} \{K_{Rf}\}^T \{K_{Rf}\} + \{K_{If}\}^T \{K_{If}\} \right]^{-1} \sum_{f=1}^{n_k} [\mathbf{Q}_{Rf}] \{K_{Rf}\}^T + [\mathbf{Q}_{If}] \{K_{If}\}^T \quad (30)$$

When all $\mathbf{A}^{ij}$ is calculated for each element of $\bar{\mathbf{Q}}(M, k)$, one obtains a fit with respect to reduced frequency k . In order to use $\bar{\mathbf{Q}}(M, k)$ in time domain simulations and frequency domain analysis, it is expressed in Laplace domain, using the relation $ik = sb/U$, as shown in Eq. (8).

$$[\bar{\mathbf{Q}}(M, s)] = [\mathbf{A}_0] + \frac{sb}{U}[\mathbf{A}_1] + \left(\frac{sb}{U}\right)^2 [\mathbf{A}_2] + \sum_{n=1}^N \frac{s[\mathbf{A}_{n+2}]}{s + \frac{U}{b}\beta_n} \quad (31)$$

GAF matrices are obtained from MSC Nastran in the aerodynamic set, namely the k-set. The matrices are then transformed to the m-set as shown in Eq. (32) and Eq. (33), where $\mathbf{G}_{ka}$ and ϕ_{am} are the spline matrix between the structural and aerodynamic grid points, and the reduced modal matrix respectively.

$$[\mathbf{Q}_{aa}^{ij}] = [\mathbf{G}_{ka}^T][\mathbf{Q}_{kk}^{ij}][\mathbf{G}_{ka}] \quad (32)$$

$$[\mathbf{Q}_{mm}^{ij}] = [\phi_{am}^T][\mathbf{Q}_{aa}^{ij}][\phi_{am}] \quad (33)$$

Then Roger's method is applied on the m-set GAF matrices to obtain $\bar{\mathbf{Q}}_{mm}(M, s)$. In Eq. (8), the number of aerodynamic lag roots is determined so that a satisfactory fit for GAF matrices is performed. Typically 2 to 4 lag roots are used in aeroelastic studies¹. The aerodynamic forcing in Laplace domain is finally expressed as in Eq. (34).

$$\{\mathbf{F}_m(s)\} = q[\bar{\mathbf{Q}}_{mm}(M, s)]\{\zeta_m(s)\} \quad (34)$$

In Figure 1, elements of $\bar{\mathbf{Q}}_{mm}$ and $\mathbf{Q}_{mm}$ are compared for each reduced frequency at which $\mathbf{Q}_{mm}$ is obtained.

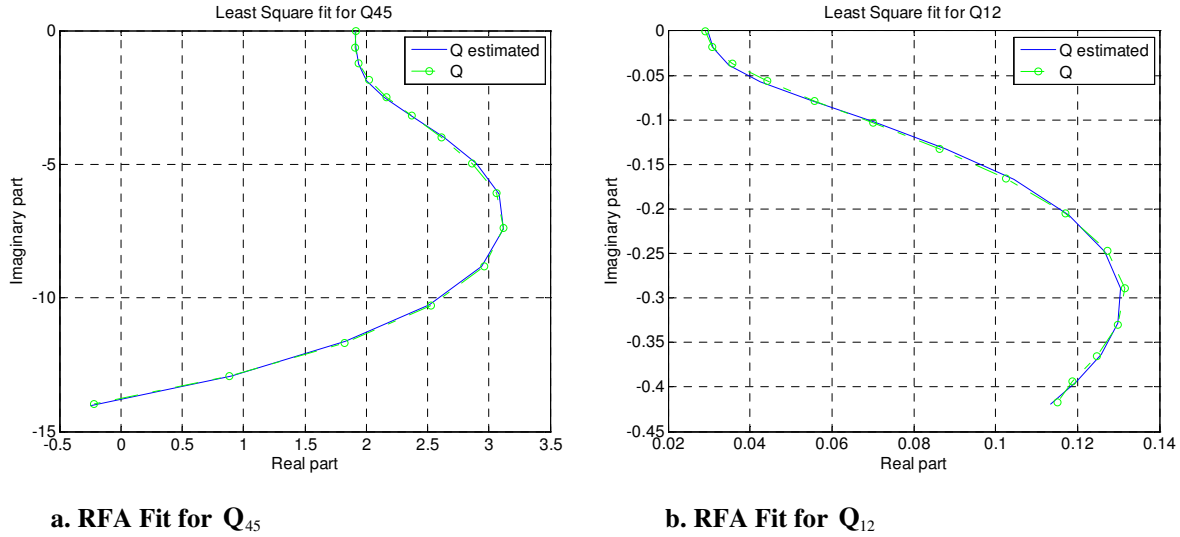


Figure 1 RFA Fit Comparison for $\mathbf{Q}_{45}$ and $\mathbf{Q}_{12}$

B. Structural Modeling

The structural connection between the missile control fin and the servo-actuation system is represented by a shaft that has rotational stiffness of 120 N.m/rad. The servo-actuation system is assumed to be composed of a DC Motor and a transmission unit. The connection between the fin and the transmission unit and the thickness distribution of the fin are depicted in Figure 2 and Figure 3 respectively.

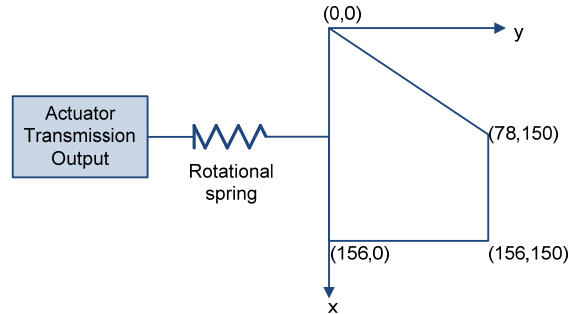


Figure 2 Transmission-Fin Connection

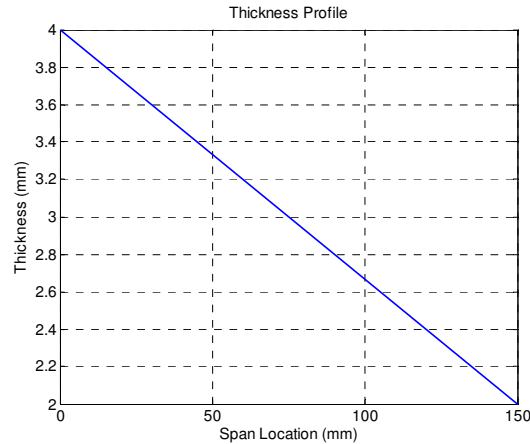
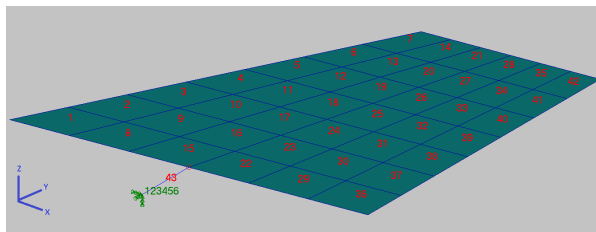
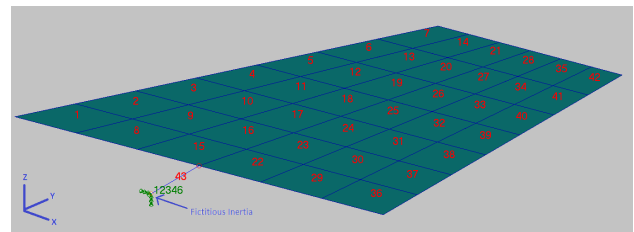


Figure 3 Thickness distribution of the control fin

IN MSC Patran, two finite element models are prepared for distinct purposes. The first finite element model (FN1) is prepared for the p-k method of flutter analysis, and normal modes analysis for the elastic modes in MSC Nastran. In this model, the fin is connected to the actuator through a shaft. This shaft is assumed to be rigid in all five degrees of freedom except for the rotational degree of freedom in the y-axis shown in Figure 4. It is assumed that the shaft reflects the resultant static stiffness of the actuator and the connecting shaft. In the y-axis, it has a stiffness of 120 N.m/rad. To model this shaft connection, a point on the ground is generated such that it is fixed in all degrees of freedom. The mid-node on the root chord of the fin is rigidly connected to this point through a multi point constraint (MPC), such that all five degrees of freedom except the rotational one in y-axis is fixed for this node. The rotational stiffness of the shaft in y-axis is modeled by a one dimensional bar element which has no mass or inertia. The model is shown in Figure 4a, where the fin structure is modeled with 42 QUAD4 elements.



a. Finite Element Model FN1



b. Finite Element Model FN2

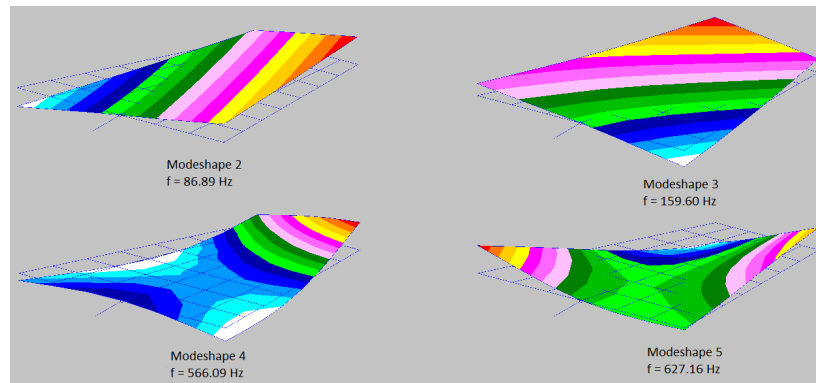
Figure 4 Finite Element Models of the Fin

A second model (FN2) is prepared, so that the rigid body mode could be obtained together with the elastic modes, without affecting the mode shapes and natural frequencies of the elastic modes. The rigid body mode is generated so that it is used for modeling the rigid control motion of the actuator. For this purpose, the boundary condition on the point on the ground is modified such that it is disconnected from the ground in the y-axis rotational degree of freedom. However, in this case, the mode shapes and the frequencies of the elastic modes are changed. To generate the needed modes in a single normal modes analysis run in MSC Nastran, a very high fictitious inertia is used in the y-axis, to mimic the fixed boundary condition at the point on the ground. The fictitious inertia is about 10^6 times greater than that of the rotational inertia of the fin. The high inertia adds a huge inertial resistance to the motion at the point where it is added. Therefore, the mode shapes and natural frequencies contain a rigid body mode, in addition to the original elastic modes and frequencies obtained from the model FN1. The mass matrix $\mathbf{M}_{mm}$ is now changed, but the additional inertia is discarded from $\mathbf{M}_{mm}$ when it is used in subsequent aeroelastic analysis in MATLAB. The model is shown in Figure 4b, together with the changed boundary condition and the node where the fictitious inertia is added. CMASS1 element is used for the added inertia, and a SUPORT statement is inserted into the Bulk Data Section of MSC Patran flutter analysis menu¹², so that the rigid body mode obtained has exactly zero natural frequency. The modes taken into account for the aeroelastic analysis are limited by an upper limit of 1000 Hz. Normal modes analysis of both models are performed by MSC Nastran, and the comparison of the natural frequencies is given in Table 1 for the 5 elastic modes besides the rigid body rotation mode.

Table 1 Comparison of natural frequencies determined by finite element models FN1 and FN2

	Model FN1 (Hz)	Model FN2 (Hz)
Rigid Body Mode	-	0.0000
Elastic Mode 1	86.8891	86.8891
Elastic Mode 2	159.6022	159.6022
Elastic Mode 3	566.0895	566.0895
Elastic Mode 4	627.1564	627.1564
Elastic Mode 5	972.5351	972.5351

As observed from Table 1, the fictitious inertia worked perfectly, and the results obtained for the elastic natural frequencies of models FN1 and FN2 are identical. Figure 5 shows the first four elastic mode shapes.

**Figure 5 Elastic Modeshapes 1-to-4**

An additional normal modes analysis is carried out in MATLAB, so that the matrices, which are extracted from MSC Nastran using the DMAP codes¹³ inserted, are verified. In MATLAB, the eigenvalue problem solver function 'eig'¹⁷ is used to obtain natural frequencies and mode shapes. For the model FN2, natural frequencies obtained from MATLAB are presented together with the results of MSC Nastran in Table 2.

Table 2 Natural Frequency Comparison of MSC Nastran and MATLAB Modal Analyses

	Model FN2 (Hz) Nastran Output	Model FN2 (Hz) MATLAB Output
Rigid Body Mode	0.0000	1.52691E-07
Elastic Mode 1	86.8891	86.8891
Elastic Mode 2	159.6022	159.6022
Elastic Mode 3	566.0895	566.0895
Elastic Mode 4	627.1564	627.1564
Elastic Mode 5	972.5351	972.5351

Since the modes are identical, the stiffness and mass matrices $\mathbf{K}_{mm}$ and $\mathbf{M}_{mm}$ obtained from MSC Nastran and implemented in MATLAB are used in further aeroelastic analysis with confidence. Note that the modal damping matrix $\mathbf{C}_{mm}$ can be set to zero for the sake of simplicity, as its effect on the response is usually insignificant when aerodynamic damping forces are present.

C. Aerodynamic Model

Aerodynamic modeling of aircraft structures, necessitates a very careful treatment of the underlying physical phenomena. The operating conditions of the missile control fin, such as given in Table 3 should be considered, in order to make physically sound assumptions about the flow.

Table 3 Flight Conditions of the Missile Control Fin

	Minimum	Maximum
Range of Altitudes	0 m	5000 m
Range of Mach Numbers	0.4	0.6
Range of Incidence Angle	-15°	+15°

The flight conditions given in Table 3 implies that the flow around the control fin is subsonic and compressible. Furthermore, the aerodynamics can be assumed to be linear in the incidence angle region that is specified¹⁴. An additional assumption, that the flow is inviscid leads to the well known DLM for unsteady flow, where the final assumption is justified by the fact that viscous effects have little effect on the lift on the fin in the relatively small incidence region. Aerodynamic modeling of the missile control fin can therefore be carried out in MSC Flightloads and Dynamics, with the DLM, for obtaining the unsteady GAF matrices for a predefined region of reduced frequency at certain Mach numbers. The GAF matrices $\mathbf{Q}_{mm}$ are then given as input to Roger's RFA method, so that the aerodynamic forces are represented in the continuous frequency domain. To verify the GAF matrices' implementation to the MATLAB codes developed, flutter analysis in both MSC Nastran and MATLAB are performed.

The Doublet Lattice Modeling of the missile control fin aerodynamics requires certain rules to be followed in MSC Flightloads and Dynamics. Two of them significantly affect the accuracy of aerodynamic forcing and subsequent aeroelastic analysis¹². There should be a minimum of 15 boxes (panels) per wavelength for each chord strip. Not less than four boxes per chord should be used. Furthermore, boxes (panels) should maintain an aspect ratio of less than 3 in the default Doublet-Lattice formulation. The parameter 'boxes per wavelength' is defined as $BPW = U_{\min} / f_{\max} \Delta x$ where, Δx is the chord length of a single panel of the aerodynamic surface at a strip. $U_{\min}$ is the minimum airspeed, and $f_{\max}$ is the maximum structural frequency that are used in the analysis. Panel discretization of the fin surface is shown in Figure 6a together with the 'boxes per wavelength' criterion and in Figure 6b together with the 'aspect ratio' criterion. For the model obtained, both criteria are satisfied.

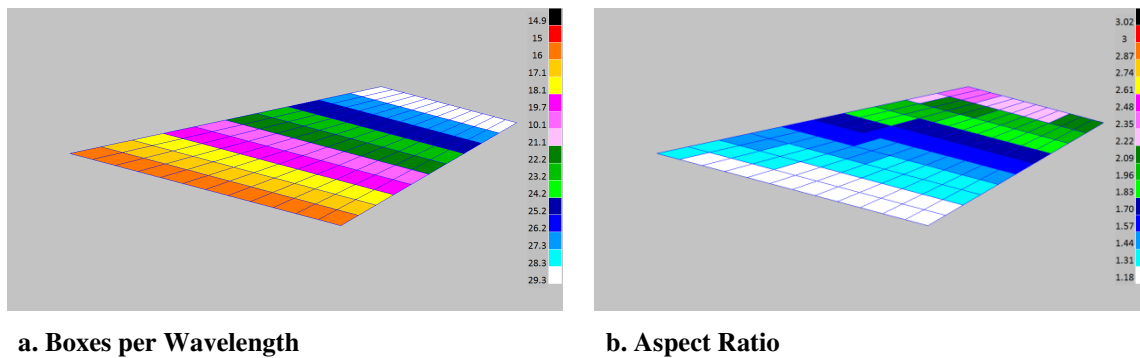


Figure 6 Aerodynamic Mesh

Figure 7 shows the DMAP code inserted to the Executive Control Deck to output the GAF Matrices at different reduced frequencies and at a single Mach number to a text file. This text file is read by MATLAB with an interfacing code developed. The GAF matrices are output in the k-set, which is the aerodynamic grid point set in MSC Nastran¹⁸.

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in the locus of roots diagram, by gradually increasing the free stream speed in q . $\mathbf{A}_{sys}$ has many eigenvalues which involve information about the dynamics of both the aeroelastic generalized coordinates and the aerodynamic lag states. When searching for flutter, one is interested in the stability of the generalized coordinates. Note that the transfer function for the aerodynamic lag states is given in Eq. (8), and the poles of that transfer function is selected to be on the left hand side of the s-plane, so that the resulting aerodynamic lags do not explicitly bring instability to the system. Then only the eigenvalues corresponding to the generalized coordinates are plotted on the root locus. The only aeroelastic system matrix that is subjected to the root locus analysis in this study is constructed by utilizing the GAF database at Mach 0.75, at which the flutter analysis was carried out with the p-k method. The eigenvalues corresponding to the generalized coordinates are obtained by using the MATLAB command 'eig' for $\mathbf{A}_{sys}$ at selected free stream speeds which range from 180 m/s to 340 m/s, with 10 m/s intervals. System matrix of a damped system gives complex eigenvalues λ , such that:

$$\lambda_j = -\xi_j \omega_j + i \omega_j \sqrt{1 - \xi_j^2} \quad (35)$$

After each λ_j is obtained from eigenvalue analysis, ξ_j effective damping ratio ξ_j and ω_j undamped natural frequency ω_j are obtained for each aeroelastic mode, where the subscript 'j' denotes an aeroelastic mode. The locus of roots for all aeroelastic modes together is given in Figure 9.

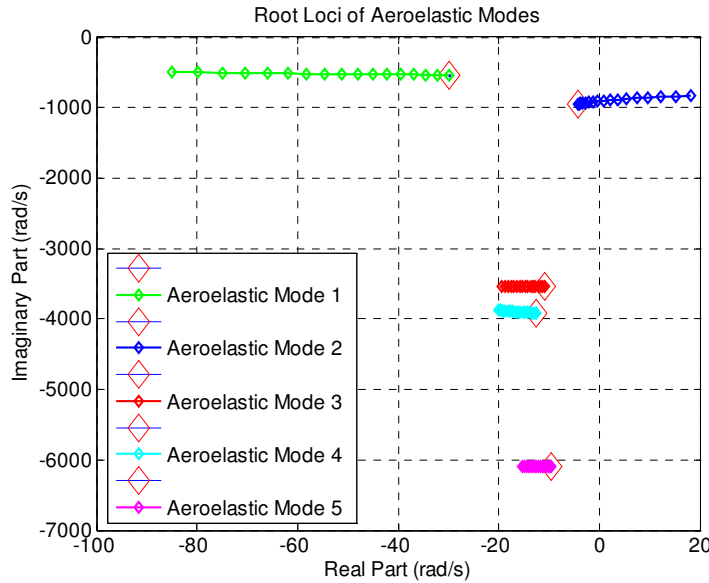


Figure 9 Locus of Aeroelastic Roots

Since the real part of the aeroelastic roots have positive values as the airspeed increases, instability is found in the second aeroelastic mode. The flutter speed is found to be at 253.11 m/s, at which the damping ratio of the second aeroelastic mode is zero. Corresponding to the flutter speed, flutter frequency is obtained as 144.1 Hz. As observed from Figure 9, the only aeroelastic mode that has a positive real part is the second mode. This result is equivalent to the one obtained in p-k method analysis. For the modes analyzed, no mechanism for instability is observed in modes other than the second aeroelastic mode.

4. Time Domain Solution:

In the present study, time domain solution of flutter is also carried out, so that the aeroelastic system matrix that will be used for the state space analysis is validated. For this purpose, the aeroelastic system matrix constructed with the GAF Database at Mach 0.75 is used. By assigning a unit initial condition to the generalized coordinate of the second aeroelastic mode, time response of the generalized coordinates of the open loop aeroelastic system is simulated with 1 m/s airspeed intervals. Eq. (36) is integrated with a variable step solver (ode45) built in MATLAB, to obtain the time response of generalized coordinates for the quarter of a second.

$$\{\dot{z}(t)\} = [\mathbf{A}_{sys}] \{z(t)\} \quad (36)$$

In Eq. (21), the state vector includes aeroelastic generalized coordinates. The initial condition for $z(t)$ is given such that all elements of $z(0)$ is zero except the second aeroelastic generalized coordinate. That initial value is set to unity. Time responses of the generalized coordinate corresponding to the second aeroelastic mode at selected free stream speeds are presented in Figure 10, Figure 11 and Figure 12. From Figs. 10-12, it is seen that flutter speed is approximately 254 m/s, because at 254 m/s, amplitude of oscillations neither decay nor increase.

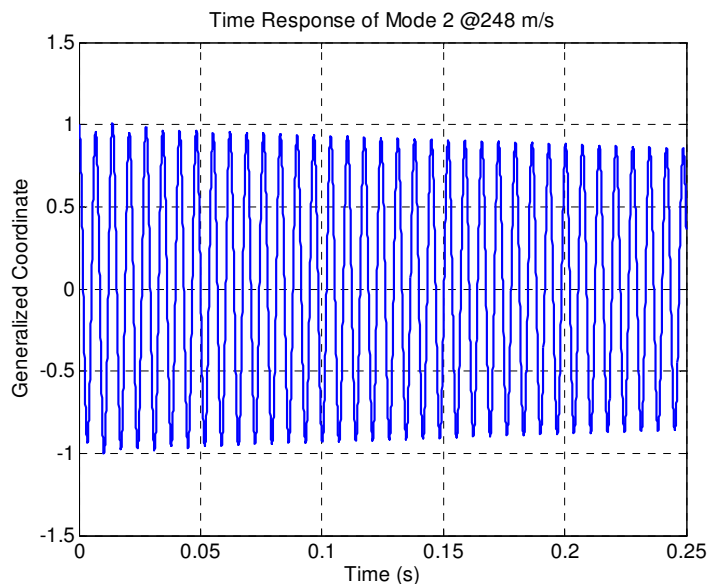


Figure 10 Time Simulation of the Second Aeroelastic Generalized Coordinate at 248 m/s

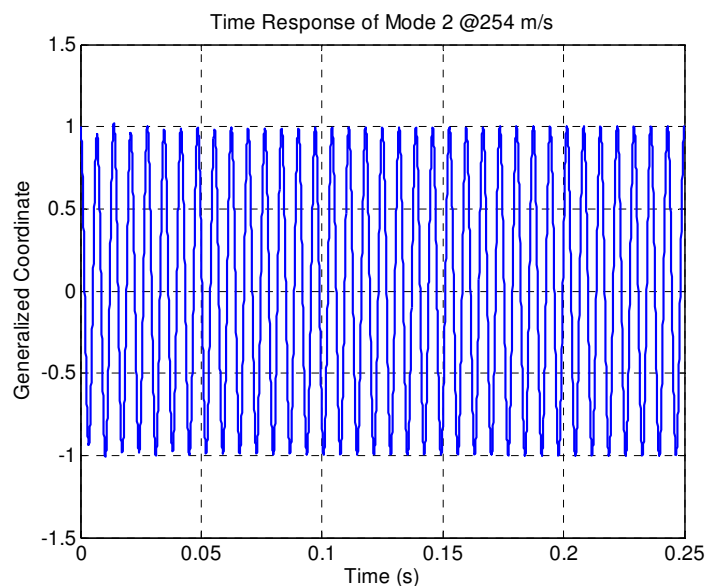


Figure 11 Time Simulation of the Second Aeroelastic Generalized Coordinate at 254 m/s

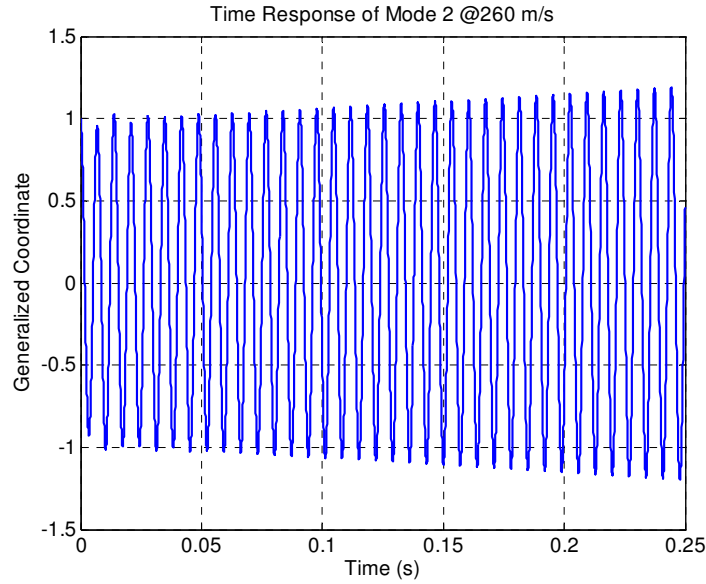


Figure 12 Time Simulation of the Second Aeroelastic Generalized Coordinate at 260 m/s

5. Comparison of Flutter Solutions

The flutter speeds obtained by three different methods at the sea level are given in Table 5 for comparison.

Table 5 Flutter Speed Comparison

	P-k Method (MSC Nastran)	Root Locus Method (MATLAB)	Time Domain Solution (MATLAB)
Altitude (m)	0	0	0
Mach	0.75	0.75	0.75
Air Density (kg/m^3)	1.226	1.226	1.226
Speed of Sound	340.3	340.3	340.3
Flutter Point	Aeroelastic Mode 3	Aeroelastic Mode 3	Aeroelastic Mode 3
Flutter Speed (m/s)	251.12	253.11	254
Flutter Frequency (Hz)	143.55	144.1	-
U^{eq} (m/s)	251.12	253.11	254
M^{eq}	0.74	0.74	0.74

The flutter results presented in Table 5, are considered to be almost identical. Therefore it is concluded that the RFA fitting process and the aeroelastic system matrix construction is valid. Note that U^{eq} and M^{eq} represents the equivalent airspeed and equivalent Mach number, which should match the Mach number used in the analysis.

III. Actuator and Control System Modeling

Hydraulic, pneumatic and electromechanical servo actuation systems have distinctive dynamic characteristics, which will make the response of the aeroservoelastic system differ in terms of both stability and performance. An electromechanical servo-actuator system model is integrated to the aeroelastic system in this work, to study the aeroservoelastic dynamics. The electromechanical servo-actuation system is made up of a DC Motor, a transmission unit and a PD Controller. In Figure 13, a generic block diagram of a servo-actuation system is shown.

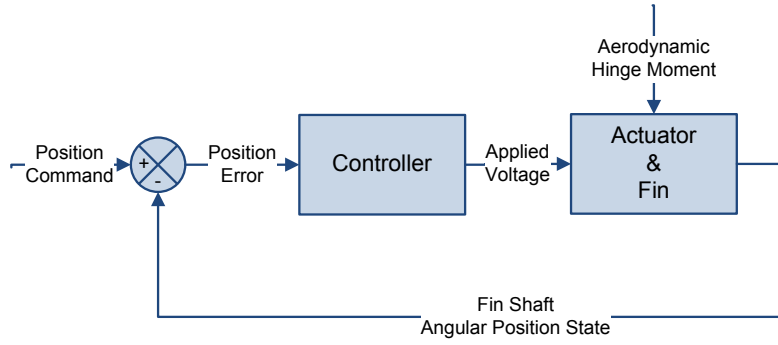


Figure 13 Block Diagram of the Servo-Actuation System

Linear transfer functions for the servo-actuation system are presented in this section. These transfer functions are used for Root Locus¹⁵ design of the PD controller. Nonlinear parameters such as dry friction and free play in mechanical linkages are neglected. On the other hand, the effect of the power limit nonlinearity of the DC Motor is modeled in MATLAB Simulink for performance analysis.

The transmission unit is represented with a specific speed reduction ratio and efficiency, which is modeled by Eq. (37) to Eq. (39):

$$\theta_m = \theta_{fin} N_{tr} \quad (37)$$

$$J_{total @ motor} = J_{rotor} + J_{gear @ motor} + \frac{J_{fin}}{N_{tr}^2 \eta_{tr}} \quad (38)$$

$$T_{hinge @ motor} = \frac{T_{hinge @ fin}}{N_{tr} \eta_{tr}} \quad (39)$$

where the parameters in Eq. (37) to Eq. (39) are defined as:

J_{rotor}	: Rotor Inertia (kg.m ²)	θ_m	: Angular Position of Motor Shaft
$J_{gear @ motor}$	: Gear Inertia (kg.m ²)	θ_{fin}	: Angular Position of Fin Shaft
$T_{hinge @ motor}$	: Hinge Moment on the Motor (N.m)	N_{tr}	: Combined Speed Reduction Ratio
$T_{hinge @ fin}$	: Hinge Moment on the Fin (N.m)	η_{tr}	: Combined Efficiency

J_{rotor} and $J_{gear @ motor}$ are obtained from DC motor catalogs and J_{fin} is obtained from the MSC Patran finite element model.

Dynamic characteristics of a DC Motor are often given in motor catalogs in terms of constant parameters. As long as the DC Motor is used according to the operating conditions that are specified in the catalog such as nominal voltage and max load torque, these parameters are valid for the motor's entire lifecycle. The parameters are listed below:

V	: Nominal Voltage (Volts)	R	: Armature Resistance (Ohms)
K_t	: Torque Constant (N.m / Amperes)	L	: Armature Inductance (Henries)
K_b	: Back EMF constant (Volts.s / radians)	c	: Viscous Damping Coefficient (N.m.s/radians)

When the fin and the transmission units are assumed to be rigid, the output of the transmission unit is equivalent to the output of the fin. The angular position response of the unity feedback servo-actuation system with a PD controller is represented in Eq. (40).

$$\frac{\theta_{fin}}{\theta_{com}} = \frac{K_D K_t s + K_P K_t}{N J L s^3 + N (J_{total @ motor} R + c L) s^2 + (N (K_b K_t + c R) + K_t K_D) s + K_P K_t} \quad (40)$$

where θ_{com} is the commanded fin deflection, which is equivalent to $\zeta_c(t)$.

Note that the aerodynamic hinge moment is not included in Eq. (40), it is treated as a disturbance to the servo-actuation system. The PD controller of the servo-actuation system is designed using the Root Locus method¹⁵ by neglecting the elasticity of the fin and unsteady aerodynamic moment on the fin, as if the fin is rigid and the aerodynamics is steady. The performance requirements of the control fin response, which drive the PD controller design are given in Table 6.

Table 6 Performance Requirements of the Control Fin Response

Settling Time up to 15° Angular Position Command	50 ms
Steady State Angular Position Error	2%
Maximum Load Torque	6 N.m
Minimum Angular Speed at Maximum Load Torque	300°/s

The two primary performance requirements that lead to the determination of the servo-actuation system components, the DC Motor and the transmission unit, are the required torque and the required angular velocity at the fin. The total maximum load torque on the servo-actuator comes from the aerodynamic moment and the acceleration torque due to the inertia of the fin. The stall torque of the actuator should be higher than the resultant of the two. In general, the inertial torque is small relative to the aerodynamic torque, thus neglected in the control design. On the other hand, the size constraints on the actuator unit are demanding in missile applications. A missile is required to be light and compact as possible for any given mission concept, to reduce the costs and improve the dynamic performance. Therefore, strict size constraints are set up on all missile subsystems, including the servo-actuation system. In this case, the DC motor is selected according to a diameter constraint (<28 mm). The DC Motor Specifications are given in Table 7. Since the DC motor is specified, a transmission unit that gives the required maximum torque of 6 N.m. is to be selected. No reference to any specific brand or type of transmission is given here. A transmission unit just capable of producing the necessary amount of torque is assumed with following characteristics. The DC motor specifications are given in Table 7. The transmission properties assumed are : $N_{ir} = 66$ $\eta_{ir} = 70\%$ $J_{ir} = 0.5 \times 10^{-7} \text{ kg.m}^2$.

Table 7 DC Motor Specifications

Faulhaber 2642 024CR		
Diameter	26	Mm
Length	42	Mm
Voltage	24	Volts
Terminal Resistance	5,78	Ohms
Rotor Inductance	550	uH
Torque Constant	34,6	mNm/A
Rotor Inertia	11	g.cm ²

Note that a current limit is implemented through limiting the applied voltage between $(-I_{lim}R, I_{lim}R)$, in addition to a 24V limit for power limited nonlinear simulations.

E. PD Controller Design

The controller design is carried out in the s-plane with the root locus method. The poles of the open loop actuator transfer function are $s_1 = 0$, $s_2 = -168.80$, and $s_3 = -10340.28$. The root locus method is a good candidate for systems having such dominant pair of poles. Since the aerodynamic hinge moment is assumed to be an external disturbance while deriving the actuator transfer function, the robustness of the closed loop servo-actuation system to the aerodynamic hinge moment is assessed in PD controller design by analysis. No integral control action is preferred, because the open loop actuator transfer function is inherently a Type I system¹⁶, which has zero steady state error. The root locus of the open loop actuator transfer function without the controller near the origin is shown in Figure 14.

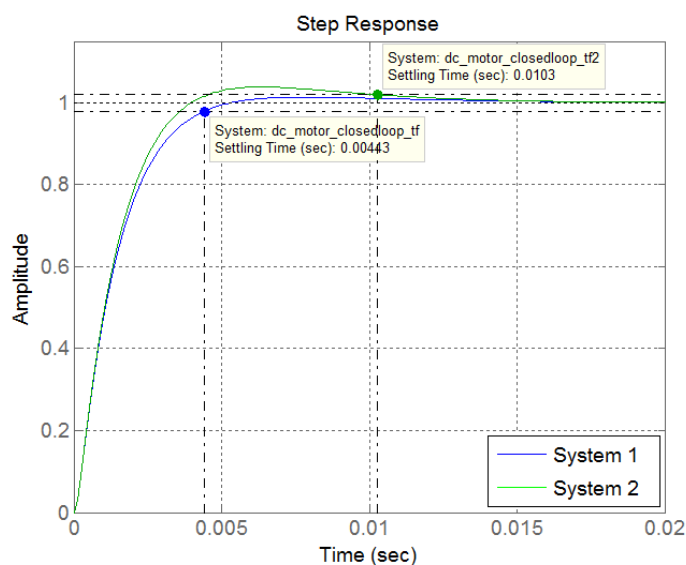


Figure 16 Step Response of Alternative Linear Closed Loop Systems

According to Figure 16, both systems are quite fast, the second one being faster, because the location of the zero is further away on the negative real axis. In the nonlinear system with power source limits, the system that performs faster in linear analysis may be advantageous. Also, it is possible to increase the controller gain K_c further for the second system before reaching the limit $s_{\min}$. The second system with power source limits is simulated to obtain the step response to 14° angular position command and the response is given in Figure 17. The steady aerodynamic hinge moment varying with angular position is applied to the nonlinear system, to check the closed loop system response under anticipated loading conditions.

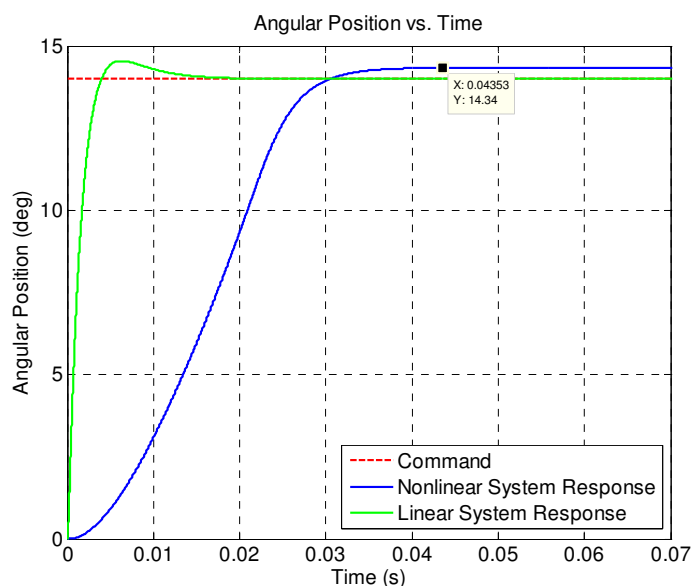


Figure 17 Step Response of System 2 to 14° Command (Linear vs. Power Limited Nonlinear)

Although the system has no overshoot, the steady state error is not acceptable according to the specification which is given as 2%. To meet the 2% requirement the PD controller is tuned such that $K_c = 10$, $K_p = 275$ and

$K_D = 1$, and a third system is formed. The step response of System 3 to 14° angular position command is given in Figure 18.

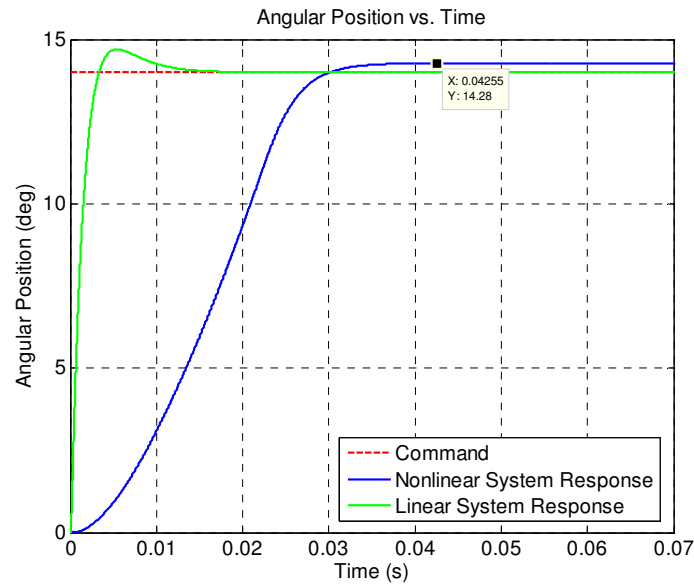


Figure 18 Step Response of System 3 to 14° Command (Linear vs. Power Limited Nonlinear)

Note that, the final system meets the steady state error requirement, settling time requirement and overshoot requirement all together. The steady state error requirement is more demanding than the others, therefore the system turned out to be faster than required. The angular position response to 0° command when the initial condition is 15° , is presented in Figure 19.

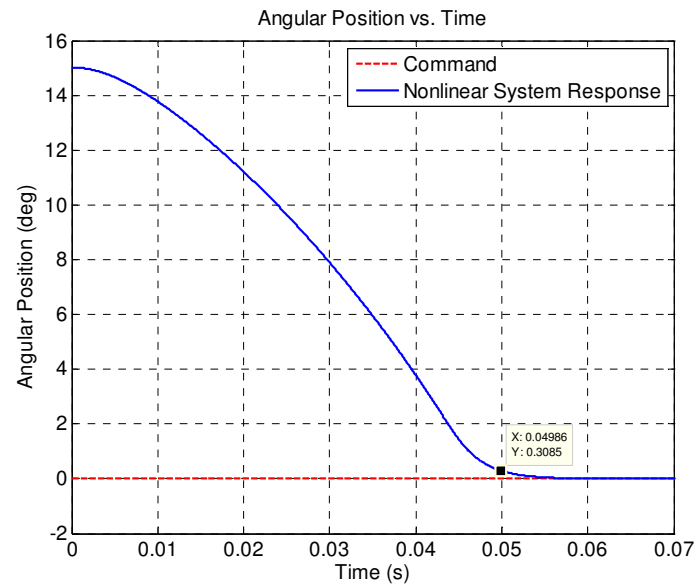


Figure 19 Angular Position Response of the Power Limited Nonlinear System

IV. Aeroservoelastic Model Integration

The mathematical models of servo-actuator and the aeroelastic systems should be integrated, so that the interaction between the control system dynamics and the aeroelastic fin could be studied. By utilizing the MATLAB

command 'lft'¹⁷ for the generalized feedback interconnection of the actuator and the aeroelastic fin, the linear transfer function of the final aeroservoelastic plant to be controlled is obtained. Figure 20 describes the feedback path for the hinge moment due to inertial and aerodynamic forcing agents.

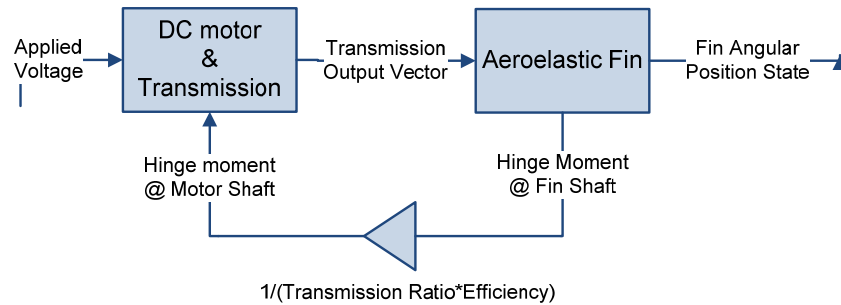


Figure 20 Open Loop Actuator & Aeroelastic Fin System

Two types of aeroservoelastic model topology are used for the frequency and the time domain analysis. The topologies differ in terms of the feedback signal of the controller. Aeroservoelastic Model 1 (AEM1) is assumed to have perfect sensor that measures the motor shaft position. The topology of Aeroservoelastic Model 1 is described in Figure 21.

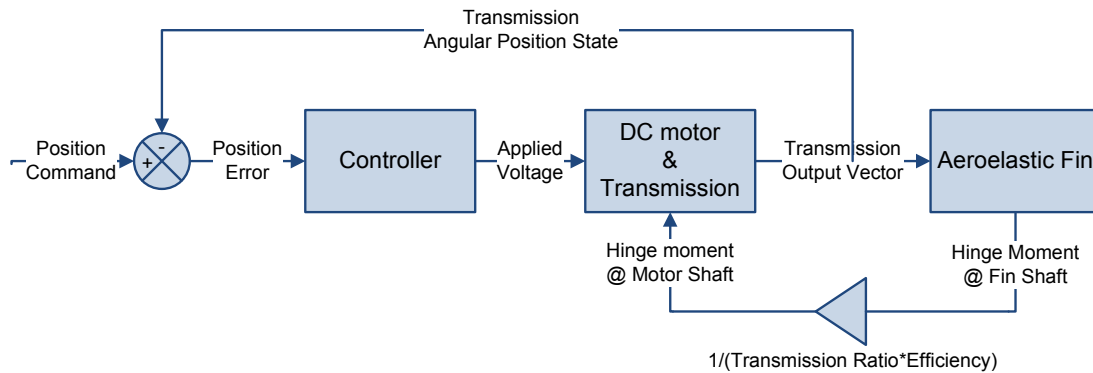


Figure 21 Aeroservoelastic Model 1 (AEM1) Topology

Aeroservoelastic Model 2 (AEM2) is assumed to have a perfect feedback sensor at the fin shaft, that directly measures the angular deflection at the fin root.

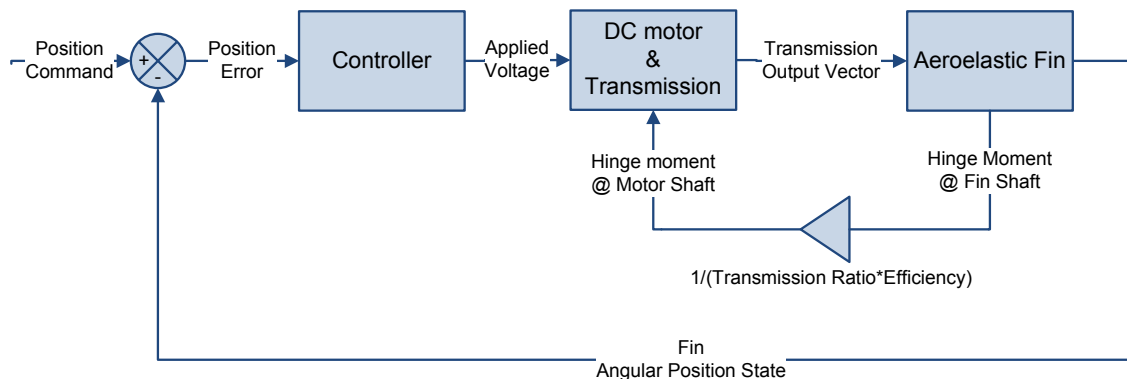


Figure 22 Aeroservoelastic Model 2 (AEM2) Topology

Utilizing the linear models AEM1 and AEM2 time domain and frequency domain analysis could be performed. In addition, the effect of the power limit nonlinearity is introduced by saturating the applied voltage by 24V and the

current by 10A. The overall aeroservoelastic system is modeled also in MATLAB Simulink, so that the nonlinearity could be implemented and simulated.

In order to study the aeroelastic effects with different feedback mechanisms, an initial frequency domain analysis is performed with the forward path transfer functions obtained by cascading the PD controller to three open loop systems; the rigid fin – steady aerodynamics system, AEM1 and AEM2. The forward path transfer functions relate the angular position error to the angular position output of the related feedback mechanism. They are named FWTF1, FWTF2 and FWTF3 respectively. The Bode diagram obtained from the analysis is shown in Figure 23. The linear systems revealed that AEM2 is unstable, since the phase crosses -180° before the gain falls below 0 dB. Other systems are stable.

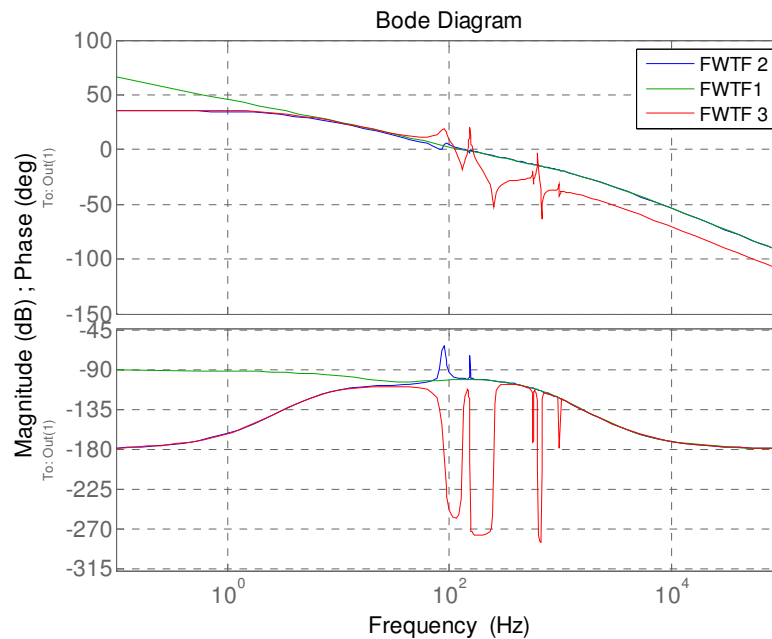


Figure 23 Comparison of Magnitude and Phase Responses of Forward Path Transfer Functions

V. Numerical Simulations of the Aeroservoelastic System

In order to reveal the interaction between control system dynamics and the aeroelastic fin, linear and power source limited nonlinear time domain simulations are performed with both AEM1 and AEM2.

Figure 24 shows the unit step response of AEM1, with and without the effect of power limit, which is specified as 24 V voltage limit and 10 A current limit. According to Figure 24, the nonlinear, power limited response still satisfies the settling time and steady state error requirements given in Table 6. Little oscillation is observed on the transmission output angular position response, which reflects the motor shaft output divided by the combined transmission ratio. Since the linear system behaves faster, there is inevitably an overshoot and more oscillation in the linear response. The transmission output angular position turned out to be satisfactory in terms of the time domain requirements.

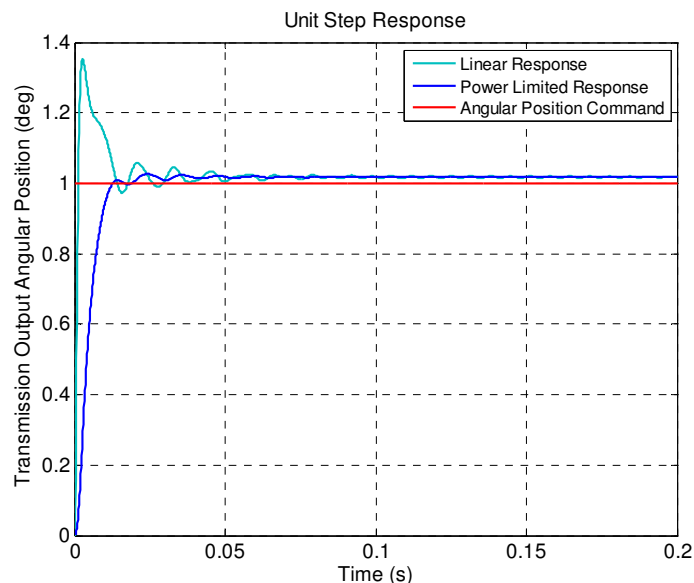


Figure 24 Time History of Transmission Output of AEM1 for Unit Step Input (Linear and Power Limited Nonlinear Models)

In Figure 25, the angular positions at various elastic degrees of freedom are given for a unit step response of AEM1. The lower left corner and the upper right corner are outer corners of the 1st and the 42nd QUAD4 elements given in Figure 26.

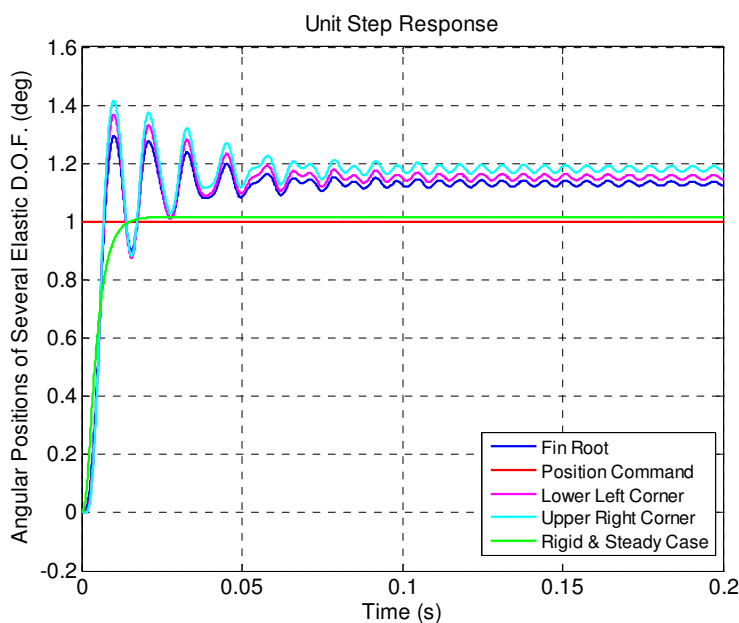


Figure 25 Time History of Various Elastic D.O.F of AEM1 and Rigid Fin for Unit Step Input (Power Limited Nonlinear Model)

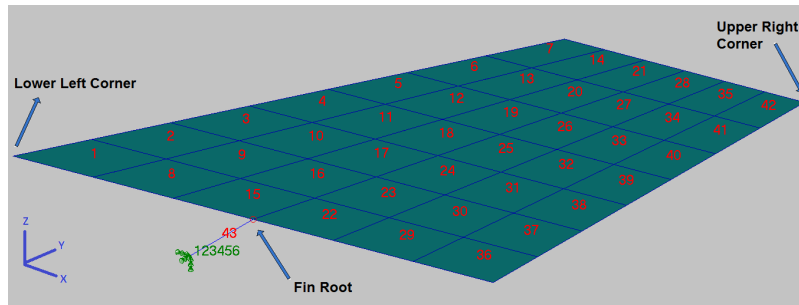


Figure 26 Fin Positions Demonstrated in Figure 25

In Figure 25, it is observed that there is an overshoot of 20% to 40% at different locations on the fin. The response converges to steady oscillation with static elastic deflection amplitude of about 15% of the input. Therefore, it is actually not possible to say that the steady state error requirement is fulfilled when the flexibility of the fin is taken into account. A more qualified approach to this requirement would be to question the total steady lift generated by the control fin at this static elastic deflection, so that the amount of control force provided to the missile could be verified. The green line in Figure 25 is the time history of the servo-actuation system model involving a rigid fin and only steady aerodynamic moment. The deviation of the elastic response from the rigid fin in terms of steady state error and damping behavior has shown that the fidelity of the rigid fin – steady aerodynamics modeling is unsatisfactory.

The contribution of all five aeroelastic modes to the response can be observed by inspecting the aeroelastic generalized coordinates' time history in Figure 27. Since the first 50 ms of motion is dominated by the transient behavior of the servo-actuation system, the first aeroelastic mode is excited. As a result, its contribution to the oscillatory response is dominant in the transient region. After reaching a steady state value, the oscillatory contribution of the first aeroelastic mode dies out, and the oscillatory response becomes dominated by the second aeroelastic mode. The oscillations of the second mode dies out rather slower, since this mode is the one at which the flutter is expected. The steady state values of aeroelastic generalized coordinates converging to nonzero values imply the existence of the static elastic deflection.

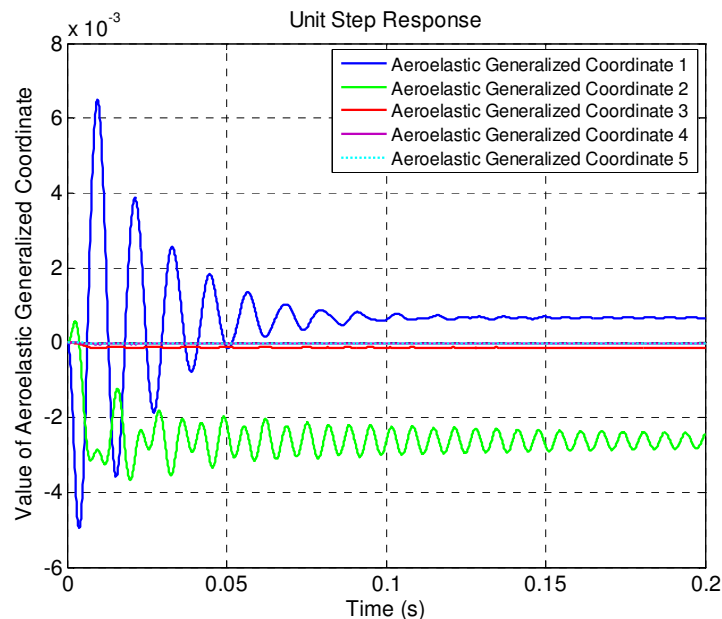


Figure 27 Time History of Aeroelastic Generalized Coordinates of AEM1 for Unit Step Input (Power Limited Nonlinear Model)

The aeroelastic hinge moment history is given in Figure 28, together with the steady aerodynamic moment estimate for the same angular position response. Note that the steady state values of the two hinge moments are different due to elastic static deflection of the fin. The oscillatory nature of the load implies more wear, and more power consumption to compensate for the fluctuations.

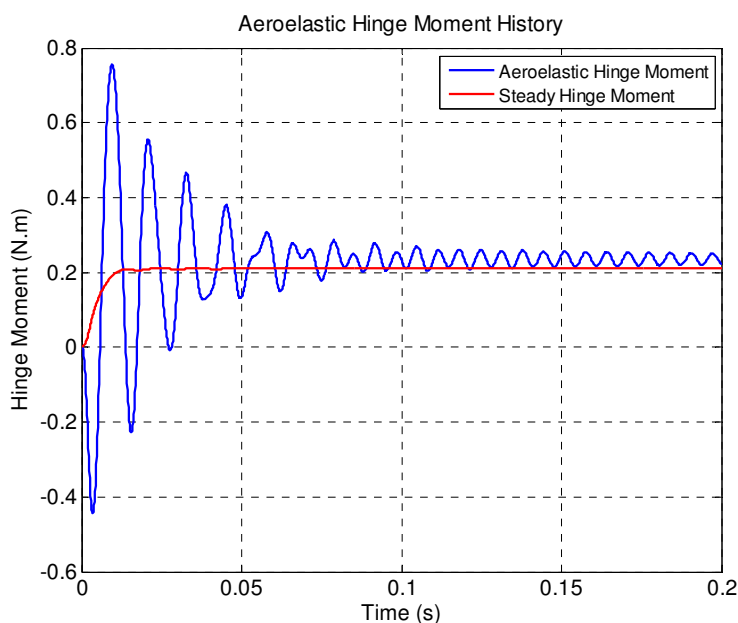


Figure 28 Time History of Hinge Moment of AEM1 for Unit Step Input (Power Limited Nonlinear Model)

The response at the transmission output, to a $+15^\circ$ step input is simulated and shown in Figure 29, so that the effect of the power limit is fully observed. The nonlinear position response with power limit has very little oscillation and no overshoot, because the system slowed down at higher values of the input due to increased effect of power saturation. The nonlinearity imposed a desirable effect, reducing overshoot and oscillation, by limiting the bandwidth of the control system.

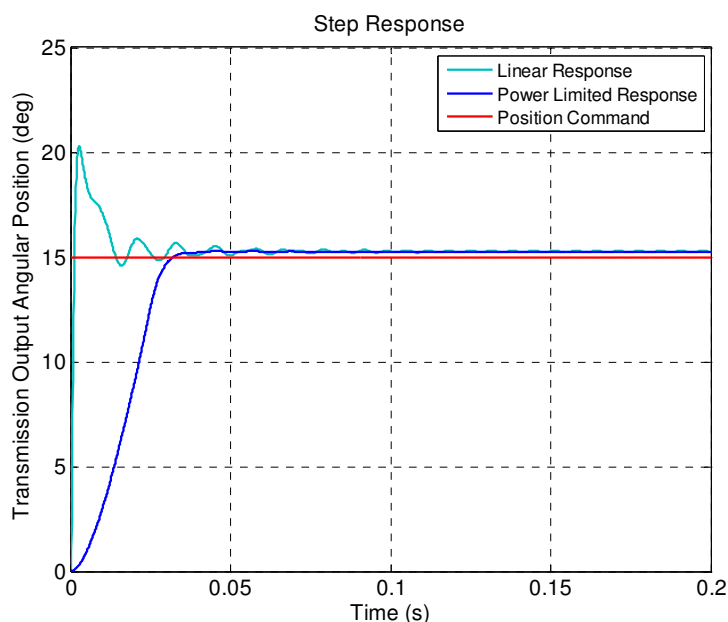


Figure 29 Time History of Transmission Output of AEM1 for $+15^\circ$ Step Input

The voltage and current time histories are given in Figure 30 and Figure 31 respectively for the unit step input and the $+15^\circ$ step input. It is observed that the voltage input of the actuator of AEM1 is saturated for a longer time, when the input is $+15^\circ$. This input dependency and the limiting nature of the system is inherently nonlinear, but seems to be desirable.

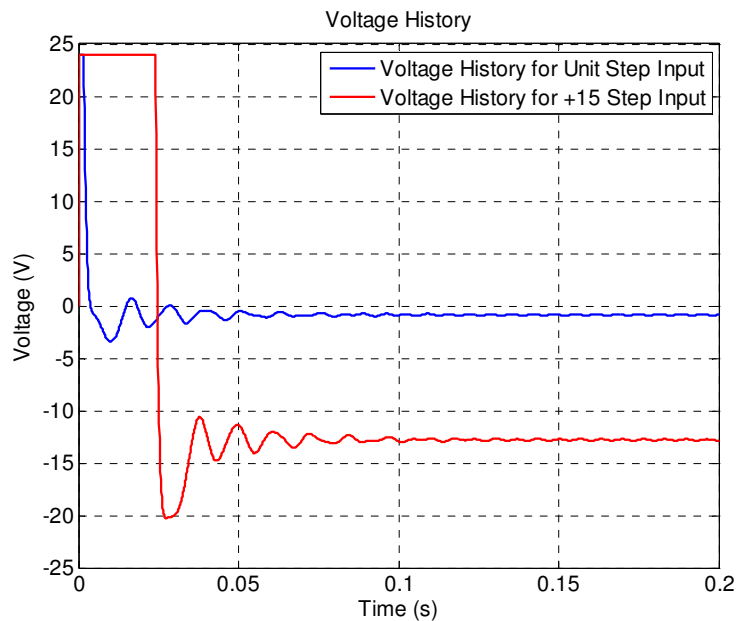


Figure 30 Time History of Voltage Input of AEM1(Power Limited Nonlinear Model)

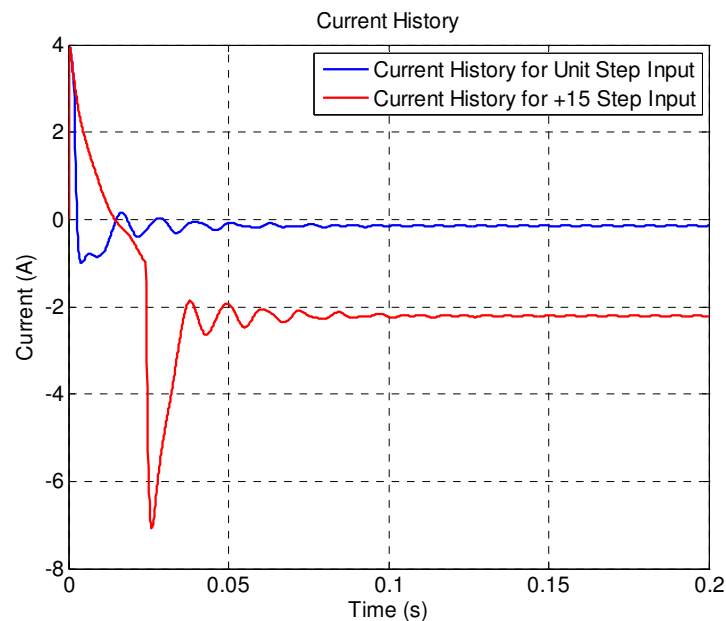


Figure 31 Time History of Voltage Input of AEM1(Power Limited Nonlinear Model)

Figure 32 shows the unit step response of AEM2, without the effect of power limit. According to Figure 32, the linear response is unstable. In this case, because the system is linear, the position response grows unboundedly and AEM2 linear model is dynamically unstable.

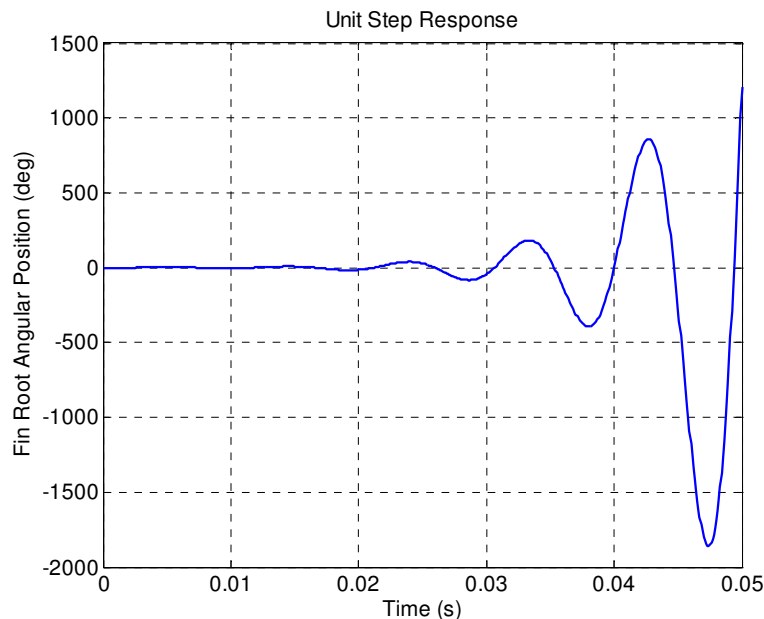


Figure 32 Time History of Fin Root of AEM2 for Unit Step Input (Linear Model)

Figure 33 shows the unit step response of AEM2 with the power limit. The response in this nonlinear case inherits the behavior of many nonlinear unstable systems, namely the limit cycle oscillation. This limit cycle oscillation actually seems to be of desirable type, since it prevents the system from going unstable, but the oscillation amplitude is clearly unacceptable. A nonlinearity may also introduce instability to a system, whereas the linearized version of the system is originally stable. This type of instability may have been observed in AEM1, but the limit on voltage is so high that it did not introduce such effect.

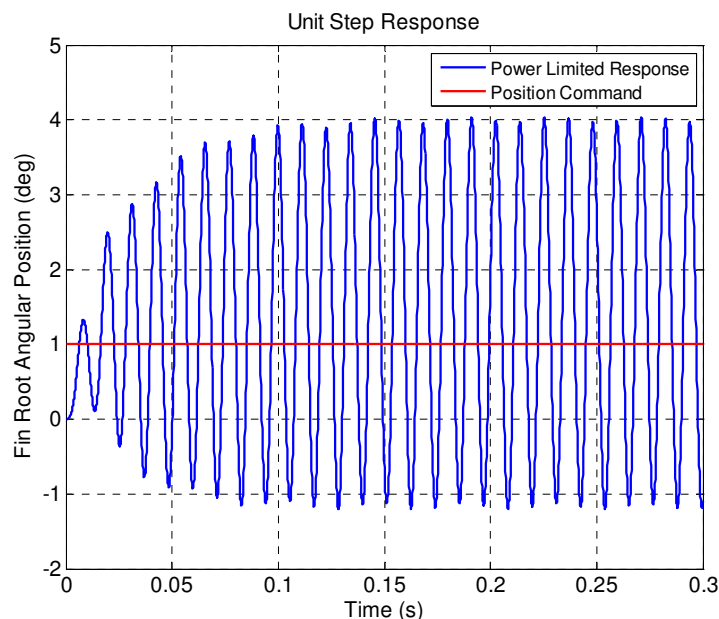


Figure 33 Time History of Fin Root of AEM2 for Unit Step Input (Power Limited Nonlinear Model)

The contribution of all five aeroelastic modes can be observed by inspecting the aeroelastic generalized coordinates' history in Figure 34. The oscillations of the first two aeroelastic modes are sustained, do not grow or

decay. The steady state values of aeroelastic generalized coordinates converging to nonzero values imply the existence of the static elastic deflection. In AEM1, second aeroelastic mode was dominant for the steady state response. On the contrary, the first aeroelastic mode seems to be more pronounced than the second aeroelastic mode in AEM2. This is due to the stronger coupling of the servo-actuator dynamics and aeroelastic dynamics in AEM2. Since the first aeroelastic mode's frequency is closer to the servo-actuation system bandwidth than the second aeroelastic mode, it dominates the oscillatory response both for the transient and for the steady state response.

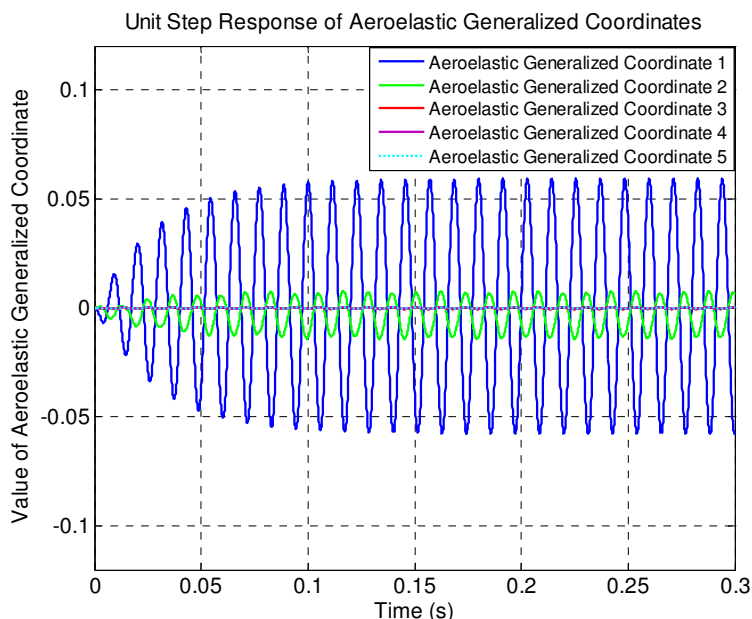


Figure 34 Time History of Aeroelastic Generalized Coordinates of AEM2 for Unit Step Input (Power Limited Nonlinear Model)

VI. Conclusion

In this study, aeroservoelastic modeling and analysis of a missile control surface which is operated and controlled by a power limited nonlinear electromechanical actuator is performed. Two different aeroservoelastic model topologies are investigated for stability, both by linearity assumption and by implementing power source limits in the servo-actuator model. It is seen that the power limited nonlinear models, were slower in response, they had lower overshoots and oscillations. The linear analysis is shown to lack fidelity in the presence of a power limit saturation by means of predicting stability and dynamic response.

The first model topology having the transmission output position as the feedback signal, named AEM1, turned out to be a stable aeroservoelastic system. The fin's static deflection and oscillatory behavior in AEM1 showed that the fin could not satisfy its 2% steady state error and damped response design requirements. It is concluded that, under the effect of aerelastic loading, the rigid fin - steady aerodynamics assumption may mislead the control system designer and result in infeasible operating conditions.

The second model topology having the fin root position as the feedback signal, named AEM2, responded to input angular position commands with limit cycle oscillations due to the voltage input limit. The oscillation amplitude turned out to be unacceptable. It is of importance that AEM2 is predicted to be an unstable system in linear analysis. It is seen that stability predictions of aeroservoelastic systems with linear tools may be insufficient for the cases where the inherent control systems push the performance boundaries to the edges by means of driving the system to power saturation.

The two position feedback paths, whose sensor dynamics are not modeled, are shown to result in drastically different behaviour by means of stability and performance. Although the controller was designed without the knowledge of aeroelastic loading, it is still concluded that the selection of the feedback path should draw great attention when dealing with systems with significant elastodynamics.

For a missile with a control fin that involves significant aeroservoelastic effects, it should be noted that there is always a possibility of interference of fin's elastic motion with missile's airframe dynamics. The control effectiveness of a missile's control channel, which specifies the tendency of the missile to get affected by control surface motion would be a key parameter for the extent of interference. In addition, the frequency spacing between the fin's aeroelastic motion and the missile's airframe dynamics would play a crucial role in the extent of interference.

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